

Fastening-Station — Standard Operating Procedure

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Audience: Maintenance Tech · Line Manager · Quality Worker
Applies to: Line-C, station 4, Fastening-Station

1. Purpose

The Fastening-Station inserts torque-controlled fasteners (bolts and screws) into the inspected weld assembly before Assembly-Robot (station 5) completes the subassembly.

2. Normal operating envelope

- Ideal cycle time: **15 s**
- Max acceptable cycle time: **19 s**
- Expected reject rate: **≤ 1 %**
- Fault probability per cycle: **0.3 %**
- Input buffer capacity: **5 parts**

3. Idle reasons

idle_reason	Meaning here	First check
Starved	Weld-Inspection (station 3) is not feeding parts.	Verify Weld-Inspection state.
Blocked	Assembly-Robot (station 5) cannot accept the part.	Verify Assembly-Robot state.

4. Faults & corrective actions

4.1 Torque-Drift

- **Symptoms in telemetry:** fault_type = "Torque-Drift". Reject codes from downstream show under-torqued or over-torqued joints.
- **Likely root cause:** torque transducer drift, or air supply pressure variation on pneumatic models.

- **Corrective action:**
 1. Run the torque calibration routine against a known torque check fixture.
 2. If drift exceeds $\pm 5\%$, replace the transducer or recalibrate.
 3. Verify air pressure regulator setpoint.
- **Restart criteria:** calibration value within $\pm 2\%$ across three setpoints.

4.2 Bit-Break

- **Symptoms in telemetry:** `fault_type = "Bit-Break"`. The spindle stalls; cycle aborts.
- **Likely root cause:** worn or fatigued driver bit, or off-axis approach.
- **Corrective action:**
 1. LOTO. Remove the broken bit.
 2. Inspect the bit holder; replace if galled.
 3. Verify spindle approach centering against the master fixture.
- **Restart criteria:** five clean drives in a row.

4.3 Feeder-Jam

- **Symptoms in telemetry:** `fault_type = "Feeder-Jam"`. Fastener-feed pickoff fails.
- **Likely root cause:** fastener bridged in the supply tube, or pickoff sensor fouled.
- **Corrective action:**
 1. LOTO. Clear the supply tube.
 2. Clean the pickoff sensor face.
 3. Verify the fastener bowl amplitude per recipe.
- **Restart criteria:** 20 fasteners delivered without alarm.

5. Preventive maintenance schedule

- **Daily:** torque verification against the check fixture.
- **Weekly:** bit holder inspection; supply-tube cleaning.
- **Monthly:** transducer calibration check.
- **Quarterly:** full pneumatic system service.

6. References

- Maintenance_Workflow_Overview.pdf
- Quality_Reject_Disposition_Policy.pdf
- Line-C_03_Weld-Inspection_NDT.pdf
- Line-C_05_Assembly-Robot_SOP.pdf